Lagrange Multipliers under the Full-Rank Condition

An End-to-End Meta-Math Walkthrough

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Abstract

The rank condition makes every vector in the linearized constraint kernel the velocity of an actual feasible curve. Stationarity along those curves forces the gradient into the constraint row space. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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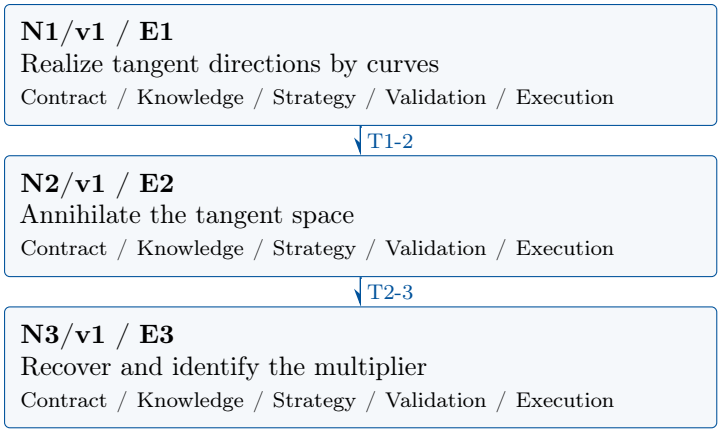
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $f : U \to \mathbb{R}$ and $h : U \to \mathbb{R}^m$ be C^1 on open $U \subset \mathbb{R}^d$, with $1 \le m < d$. Suppose $h(x_*) = 0$, $Dh(x_*)$ has rank m, and x_* is a local extremum of f on $h^{-1}(0)$. Prove there is a unique $\lambda \in \mathbb{R}^m$ with $\nabla f(x_*) = Dh(x_*)^T \lambda$. This is a necessary condition, not a general sufficient condition.*

A. Foundations and permitted imports

Import the implicit function theorem: an invertible $m \times m$ coordinate block of $Dh(x_*)$ gives a local C^1 parametrization of the zero set as a graph, whose derivative is obtained by differentiating $h = 0$. Use the chain rule and finite-dimensional linear algebra: the row space of a full-row-rank matrix A is $(\ker A)^\perp$. This identity follows from containment plus dimensions m on both sides.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. For every $v \in \ker A$ there is a C^1 feasible curve γ through x_* with $\gamma'(0) = v$.

Receiving activity. N2.1

Compatibility. E1/v1 and local constrained extremality. Check: Both signs of the curve parameter are available locally, allowing a zero derivative rather than a one-sided inequality.

T2-3: N2/E2/v1 to N3. $\nabla f(x_*) \in (\ker A)^{\perp}$.

Receiving activity. N3.1

Compatibility. E2/v1 and rank $A = m$. Check: Uniqueness of the multiplier uses injectivity of A^T, not uniqueness of the constrained optimum.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Realize tangent directions by curves

Object and role. Use the implicit graph to convert a linearized feasible direction into a genuinely feasible curve. **Local obligation:** For every $v \in \ker A$ there is a C^1 feasible curve γ through x_* with $\gamma'(0) = v$.

Inputs. $A = Dh(x_*)$ of rank m, and $v \in \ker A$.

C — Contract

Publish E1/v1 only after establishing this obligation: For every $v \in \ker A$ there is a C^1 feasible curve γ through x_* with $\gamma'(0) = v$.

K — Knowledge

$A = Dh(x_*)$ of rank m, and $v \in \ker A$. Retain the stated assumptions and import versions.

S — Strategy

Use the implicit graph to convert a linearized feasible direction into a genuinely feasible curve.

V — Validation

The invertible coordinate block is guaranteed by rank m; without it, the kernel may overestimate actual tangent directions.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 After permuting coordinates write $x = (y, z)$ with $y \in \mathbb{R}^m$ and $D_y h(x_*)$ invertible. The implicit function theorem writes nearby feasible points as $(\phi(z), z)$, with $\phi(z_*) = y_*$. Differentiating $h(\phi(z), z) = 0$ gives $D\phi(z_*) = -(D_y h)^{-1} D_z h$, all derivatives evaluated at x_*.

N1.2 Write $v = (v_y, v_z)$. The equation $Av = 0$ says $v_y = D\phi(z_*)v_z$. Hence $\gamma(t) = (\phi(z_* + tv_z), z_* + tv_z)$ is feasible for small positive and negative t, passes through x_*, and has derivative v.

Export E1/v1. For every $v \in \ker A$ there is a C^1 feasible curve γ through x_* with $\gamma'(0) = v$.

Acceptance gate. The invertible coordinate block is guaranteed by rank m; without it, the kernel may overestimate actual tangent directions.

Targeted repair. If a kernel vector is declared tangent without a feasible curve, construct it through the implicit graph and check the full-rank hypothesis. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Annihilate the tangent space

Object and role. Differentiate the objective along each two-sided feasible curve. **Local obligation:** $\nabla f(x_*) \in (\ker A)^\perp$.

Inputs. E1/v1 and local constrained extremality. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\nabla f(x_*) \in (\ker A)^\perp$.

K — Knowledge

E1/v1 and local constrained extremality. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Differentiate the objective along each two-sided feasible curve.

V — Validation

Both signs of the curve parameter are available locally, allowing a zero derivative rather than a one-sided inequality.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For each $v \in \ker A$, choose the curve from E1. The scalar C^1 function $t \mapsto f(\gamma(t))$ has a local extremum at the interior parameter value zero. Its derivative is therefore zero.

N2.2 The chain rule gives that derivative as $\langle \nabla f(x_*), v \rangle$. Since v was arbitrary in $\ker A$, the gradient annihilates that kernel.

Export E2/v1. $\nabla f(x_*) \in (\ker A)^\perp$.

Acceptance gate. Both signs of the curve parameter are available locally, allowing a zero derivative rather than a one-sided inequality.

Targeted repair. If an ambient extremum is assumed instead of a constrained one, apply stationarity only to the feasible curves already constructed. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Recover and identify the multiplier

Object and role. Identify the orthogonal complement of the kernel with the image of A^T. **Local obligation:** There is a unique λ with $\nabla f(x_*) = A^T\lambda$.

Inputs. E2/v1 and rank $A = m$. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: There is a unique λ with $\nabla f(x_*) = A^T\lambda$.

K — Knowledge

E2/v1 and rank $A = m$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Identify the orthogonal complement of the kernel with the image of A^T.

V — Validation

Uniqueness of the multiplier uses injectivity of A^T, not uniqueness of the constrained optimum.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For any λ and $v \in \ker A$, $\langle A^T\lambda, v\rangle = \langle \lambda, Av\rangle = 0$, so $\operatorname{im} A^T \subset (\ker A)^\perp$. The two spaces have dimension m, giving equality. E2 therefore supplies a multiplier.

N3.2 If $A^T\lambda = A^T\mu$, then $A^T(\lambda - \mu) = 0$. Full row rank makes A^T injective, so $\lambda = \mu$. No conclusion that every multiplier solution is an extremum has been used.

Export E3/v1. There is a unique λ with $\nabla f(x_*) = A^T\lambda$.

Acceptance gate. Uniqueness of the multiplier uses injectivity of A^T, not uniqueness of the constrained optimum.

Targeted repair. If multiplier existence is treated as sufficient for optimality, retain only the necessary condition and perform a separate second-order or convexity analysis. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Realize tangent directions by curves

1. **N1.1.** After permuting coordinates write $x = (y, z)$ with $y \in \mathbb{R}^m$ and $D_y h(x_*)$ invertible. The implicit function theorem writes nearby feasible points as $(\phi(z), z)$, with $\phi(z_*) = y_*$. Differentiating $h(\phi(z), z) = 0$ gives $D\phi(z_*) = -(D_y h)^{-1} D_z h$, all derivatives evaluated at x_*.

2. **N1.2.** Write $v = (v_y, v_z)$. The equation $Av = 0$ says $v_y = D\phi(z_*)v_z$. Hence $\gamma(t) = (\phi(z_* + tv_z), z_* + tv_z)$ is feasible for small positive and negative t, passes through x_*, and has derivative v.

B. N2 — Annihilate the tangent space

1. **N2.1.** For each $v \in \ker A$, choose the curve from E1. The scalar C^1 function $t \mapsto f(\gamma(t))$ has a local extremum at the interior parameter value zero. Its derivative is therefore zero.

2. **N2.2.** The chain rule gives that derivative as $\langle \nabla f(x_*), v \rangle$. Since v was arbitrary in $\ker A$, the gradient annihilates that kernel.

C. N3 — Recover and identify the multiplier

1. **N3.1.** For any λ and $v \in \ker A$, $\langle A^T\lambda, v\rangle = \langle \lambda, Av\rangle = 0$, so $\operatorname{im} A^T \subset (\ker A)^\perp$. The two spaces have dimension m, giving equality. E2 therefore supplies a multiplier.

2. **N3.2.** If $A^T\lambda = A^T\mu$, then $A^T(\lambda - \mu) = 0$. Full row rank makes A^T injective, so $\lambda = \mu$. No conclusion that every multiplier solution is an extremum has been used.

□

V. WORKED ORCHESTRATION

Minimize $f(x, y) = x^2 + y^2$ subject to $h(x, y) = x + y - 1 = 0$. The gradient equation is $(2x, 2y) = \lambda(1, 1)$, hence $x = y = 1/2$ and $\lambda = 1$. Substituting $y = 1 - x$ gives $f = 2(x - 1/2)^2 + 1/2$, separately verifying that this stationary point is the global minimum.

A. Boundary and hypothesis audit

If rank fails, take $f(x, y) = x$ and $h(x, y) = x^2 + y^2$. The feasible set is the single point $(0, 0)$, which is both a constrained minimum and maximum, but $\nabla f = (1, 0)$ while

$Dh = (0, 0)$, so no multiplier exists. Sufficiency also fails: for $f(x, y) = x^3$ subject to $y = 0$, the origin satisfies the multiplier equation but is not an extremum.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 inference: every vector in $\ker Dh$ is the velocity of a feasible curve, without a rank condition. The single-point constraint has $\ker Dh = \mathbb{R}^2$ at zero but only zero feasible velocities. **Repair:** invoke full rank and the implicit graph exactly as N1 does. The original theorem is preserved; the rank-free extension is rejected. Recheck tangent annihilation and multiplier existence.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Develop second-order conditions on the tangent space, explicitly separating necessary and sufficient tests.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Boyd and Vandenberghe, Convex Optimization for convexity, optimality, and descent; University of Maryland, Scientific Computing I for the numerical-analysis setting. The estimates used here are derived explicitly in §4.